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Predicting Network Growth in Finite-Sized Networks

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Layperson's Summary

It is 2026 and we exist in an interconnected world. The connectivity of our human experience is however not a new observation, in the early 20th century Hungarian poet and writer Frigyes Karinthy produced a short story centered on this idea of an interconnected world. In his 1929 short story, Chains (Láncszemek), he suggested the global population at the time of 1.5 Billion people were at most five acquaintances away from one another. This idea was popularised in John Guare's 1990 play "Six Degrees of Separation".

In 2026, however, compared with 1929 or even 1990, we exist in an interconnected world that leaves an abundance of measurable footprints. These footprints exist in the digital, in the form of data, and allow us to study this connectivity more rigorously than ever before. In 2011, using this data, an experiment was performed, finding that the average degree of separation on Twitter between two users was 3.84 [1]. In other words, researchers found the shortest chain of mutual connections typically involved fewer than four accounts. So, if written just two decades later, Guare's play would need renaming.

Amazingly, the mathematics underlying this connectivity is strikingly similar across a diverse range of systems, from molecular biology to computer science and beyond [2, 3]. When we describe such systems in terms of the connections between their constituent elements, regardless of whether they are molecules or Twitter accounts, we refer to them as networks. These networks can have very different and often complex structures. However, almost magically, many exhibit properties that can be captured and compared using the exact same mathematical equations. Such equations can offer insight into questions ranging from how biological interactions shape genetic evolution to how many acquaintances I might need before I can become friends with an Arsenal player. In this work, one such equation is created to answer the more broadly applicable question:

Given the characteristics of my network, how likely is a member within it to attract the number of connections it does?

The answer presented follows from the fact that, for a wide variety of networks, growth when new members are added is accurately described by just two processes: already well-connected members tend to attract more connections [4, 5, 6], and members with similar characteristics are more likely to connect [7, 8]. Ultimately, this work is the creation of a new equation to describe how networks grow based solely on these two processes.

To provide a more concrete example, this work focuses on networks of scientific publications (though our results would remain unchanged if we instead considered other networks, from social systems [8] to food webs [9]). In this context, one way to reframe the question above is:

Given the characteristics of an academic paper (for example, the gender of its author and its field of research), how likely is it that the paper will be referenced, say, one hundred times by other academic papers?

Given the right data (often the limiting factor), the equation created in this work can answer exactly this question. But to what degree of accuracy?

Finally, this work creates a new statistical test to assess to what extent the predictions of this equation can be relied upon. Using this test, it was found that (aside from the fact that it is not the most efficient way to make friends with Arsenal players), for the majority of simulated networks, this equation is exceedingly accurate.

Abstract

A diverse range of systems, from networks of scientific publications to social networks, exhibit growth governed by two generic processes: well-connected members attract more links (cumulative advantage), and network members with similar characteristics are more likely to connect (homophily). We develop a mathematical model that combines these mechanisms to predict how the links are distributed as such networks grow. Using asymptotic arguments, we derive a general expression for the number of links a network member is expected to receive and show how this depends on system-specific connection preferences. We also test where this asymptotic theory remains valid for finite-sized networks. To do this, we introduce a new statistical test designed for the discrete and dependent data in these networks. Using this test, simulations show the theory describes much of a finite network well, with accuracy depending strongly on the initial structure: sparse initial networks remain well described up to a larger fraction of their final network (though the extent differs when measured by network link or network member fraction) compared to denser ones, where highly linked network members attract more links than predicted. Together, these results provide a flexible model of network growth and a practical framework for determining when and where it reliably describes finite-sized networks.

1 Introduction

1.1 Motivation

It is 2026 and we exist in an interconnected world. The connectivity of human experience is however not a new observation, in the early 20th century Hungarian poet and writer Frigyes Karinthy produced a short story centered on this idea of an interconnected world. In his story, Chains (Láncszemek), he suggested the global population of 1.5 billion people were at most five acquaintances away from one another. In 2026, however, compared with 1929, we exist in an interconnected world that leaves an abundance of measurable footprints. These footprints exist in the digital, in the form of data, and allow us to study this connectivity more rigorously than ever before.

Connectivity is a property in a wide range of systems, from molecular biology to computer science [2, 3, 4]. When we describe such systems in terms of the connections between their constituent elements, be it a molecule or an Instagram account, we refer to them as networks and these elements as nodes. These networks can have complex topologies, yet share general properties that can be described using mathematical models. Such models provide insight into questions that arise across these fields, from how a user might gain more Instagram followers to how biological interactions shape genetic evolution. In this work, we produce one such model to answer the generally applicable question,

(1) *Given the characteristics of my network, how likely is a given node to attract the connections it does?*

As physicists, we aim to isolate a system’s dominant behaviours and describe them mathematically. The networks we study grow through two mechanisms: cumulative advantage [5] and homophily [7, 8]. We consider each in turn.

1.1.1 Cumulative Advantage

In 1976 Derek J. de Solla Price identified a pattern he described as “common in bibliometric matters and in many diverse social phenomena, that success seems to breed success” [5]. In this context, “success” took the form of larger volumes of a particular journal appearing year after year, creating a distribution across years from which, given a particular year, one may predict the thickness of the journal, or vice versa.

Similar empirical patterns appear in many contexts under different names, including Lotka’s law of scientific productivity [10], Bradford’s law of journal use [11], and the Pareto law of income distribution [12]. In 1968, Merton and Zuckerman termed this phenomenon the “Matthew effect” [6], naming it after the verse from the Gospel of Matthew: “For everyone who has will be given more.” Price proposed a probability distribution to describe the Matthew effect [5]. He termed this result the Cumulative Advantage Distribution, making “the general theory more available,” and moving beyond empirical laws that are “useful but not fundamental” [5].

1.1.2 Homophily

In 2001, sociologists McPherson, Smith-Lovin, and Cook described a phenomenon that “limits people’s social worlds in a way that has powerful implications for the information they receive, the attitudes they form, and the interactions they experience,” [7]. They called it homophily; the tendency for similarity to breed connection.

Homophily, in a network science context, refers to the tendency for nodes that share attributes to connect with different probabilities than nodes that do not share those attributes. For example, one might model the probability that a researcher from Imperial College London cites another researcher from the same institution differently from the probability of citing a researcher from a different university. Homophily can substantially alter the probability that a node receives a particular outcome [8]. Karimi et al. develop a network model that incorporates homophily [8]. Their model is the closest precursor to the one developed in this work. However, our model remains valid for a larger proportion of nodes within these networks.

Karimi et al. do not incorporate Price’s cumulative advantage model directly [5]. Instead, they extend the Barabási–Albert preferential attachment model [4] to include homophily. The Barabási–Albert model is essentially a rediscovery of cumulative advantage, but it describes well only nodes that have already accumulated many connections. In Matthew-effect models, highly connected nodes are more likely to acquire new links, so a small number of nodes absorb a large fraction of the total links. Most nodes therefore remain in the low-connection regime that is not accurately described by the Barabási–Albert model [4], and hence not by the Karimi et al. extension built upon it [8].

1.2 Citation Networks

Although our model is applicable to a diverse variety of networks, keeping a particular network in mind can be useful. We follow Price [5] in considering a citation network: a network of published academic papers linked by citations.

Citation networks are now stored digitally at scale. For example, the High Energy Physics Theory (hep-th) arXiv category has existed since August 1991, accumulating a large network of papers and citation links. Leskovec et al. compiled this into a citation network covering January 1993 to April 2003, containing 27,770 papers and 352,807 citations [13]. In this context, our model provides a new tool to address questions such as: How likely is a hep-th paper to receive 100 citations and how does this change depending on, say, the authors’ gender?

Homophily enters through the part of the question that depends on author attributes, in this case, gender. If similarity breeds connection, in a field that is, say, 80% male, even a slight tendency for men to cite other men can significantly affect citation patterns. Like Karimi et al., our model shows how this effect interacts with the Matthew effect (ours using cumulative advantage [5], theirs using preferential attachment [4]), whereby already highly cited hep-th papers become more likely to receive further citations.

While we focus on citation networks going forward, to reiterate, our model applies more broadly. Given the right data (often the limiting factor), it can also address questions in networks ranging from dating [8] to food webs [9], since all of these systems can be described within the same mathematical framework, graph theory.

1.3 What is a Directed Acyclic Graph?

1.3.1 Graph Definition

The term "graph" was introduced by James Joseph Sylvester in a paper published in 1878 in *Nature*, in which he drew an analogy between algebra and molecule diagrams [14]. A graph can be thought of as the underlying mathematical object, and a network as its real-world counterpart.

A graph $G = (V, E)$ is an ordered pair of two sets. The first set, V , is a set containing N elements. Each element in the set can be labeled u_i , where $i \in \{1, 2, \dots, N\}$. These elements are referred to as nodes. The second set, E , is a set of node pairs, each pair formed by choosing two nodes from the set V . These pairs are called edges [15]. For a given graph, if the set V contains N nodes, we can write,

$$V = \{u_1, u_2, \dots, u_{N-1}, u_N\}. \quad (1)$$

Since each edge can be written as a pair (u_i, u_j) of nodes, the edge set E can be written as,

$$E = \{(u_1, u_2), \dots, (u_1, u_k), \dots\}, \quad (2)$$

where $k \in \{1, \dots, N\}$.

1.3.2 A Directed Acyclic Graph

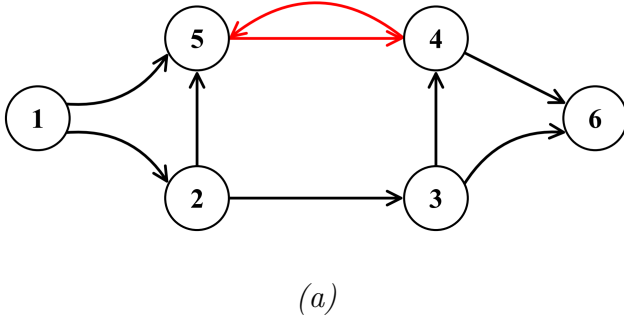
If the edges are ordered, i.e. $(u_i, u_j) \neq (u_j, u_i)$, then G is what we call a directed graph. If the edges are unordered, i.e. $(u_i, u_j) = (u_j, u_i)$, then G is an undirected graph [15]. In citation networks, direction arises from a newer paper citing an existing one, so they are naturally represented as directed graphs. Ignoring preprints and other edge cases, citations can only point backward in time and never to future papers. This temporal constraint means the network contains no cycles and is therefore acyclic (meaning the absence of cycles). Any cycle, such as $u_i \rightarrow u_j \rightarrow u_k \rightarrow u_i$, would require at least one citation to point forward in time, which is impossible in this simplified model. Therefore, once such edge cases are excluded, citation networks can be modelled as directed acyclic graphs, or DAGs for short.

To make this less abstract, a DAG is demonstrated in Figure 1. The graph representation shown in Figure 1, with nodes shown as circles and edges as arrows between them, is a common way to represent graphs and, in turn, the citation networks we wish to model.

1.3.3 In-Degree and Citation Count

The in-degree of a node, $k^{(\text{in})}$, is the number of edges pointing toward it; in a citation network, this corresponds to the number of times the paper is cited by other papers. Conversely, the out-degree, $k^{(\text{out})}$, is the number of edges pointing away from the node; in a citation network, this is the number of references the paper makes. Since we wish to model how often a paper is cited, we simply denote the in-degree as k , that is, $k^{(\text{in})} = k$. An example of in-degree and out-degree can be seen in Figure 1 (a) where node 5 has the highest in-degree with $k^{(\text{in})} = 3$ and node 6 has the lowest out-degree with $k^{(\text{out})} = 0$.

Directed Graph Citation Network



DAG Citation Network

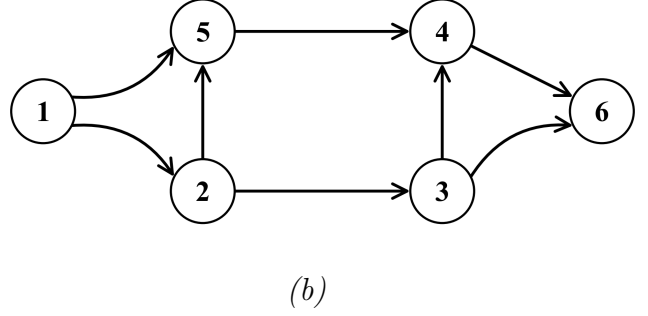


Figure 1: (a) A citation network represented as a directed graph. Each numbered circle corresponds to a publication. An arrow from paper X to paper Y indicates that X cites Y . Nodes 4 and 5, connected by reciprocated edges (shown in red), form a directed cycle. Node 5 has the highest in-degree (citation count) with $k^{(in)} = 3$. (b) A citation network with no cycles which is a directed acyclic graph (DAG). Nodes 4, 5 and 6 have the highest in-degree with $k^{(in)} = 2$.

1.4 Statistical Verification

We have now motivated the central question addressed in this thesis (question (1)), introduced the mechanisms of cumulative advantage and homophily that underpin the analysis, and developed the language of directed acyclic graphs needed to formulate the problem mathematically.

In Section 2, we will present an original derivation for this model. However, this derivation will rely on certain algebraic terms asymptotically converging to finite values in the limit that the number of nodes in our model tends to infinity.

Real-world networks, however, do not contain infinitely many nodes. As of September 2025, Instagram had 3 billion monthly active users worldwide [16]; the High Energy Physics Theory (hep-th) arXiv dataset contains 27,770 papers [13]; and an empirical food web for Little Rock Lake was reported to contain 181 taxa, where a taxon denotes a biological unit such as a species or a group of species [17]. Clearly, none of these directed acyclic graph (DAG) networks contain an infinite number of nodes.

So, this raises a question as fundamental to this thesis as question (1):

(2) *How accurate is our newly derived theoretical model at describing finite-sized networks?*

There is active debate on how best to answer this question for models similar to our own [18, 19]. In Section 4, we argue that answering Question (2) for our model requires a new goodness-of-fit test, which we provide in Section 5. The resulting test can be used to assess both our own and similar models when used to describe finite-sized networks (e.g. [4, 5, 8]). More broadly, it can be used for models describing heavy-tailed data, where extreme events remain more likely compared with exponentially decaying probability distributions [20]. One example is power law models of earthquake statistics, such as the Gutenberg–Richter law [21].

2 Theoretical Result Derivation

2.1 Network Model

This derivation is a new theoretical result for the probability that a node in a directed acyclic graph (DAG), designed to model real-world networks, has an in-degree of k at time t . In this graph, to model the different behaviors of different groups in real-world networks, nodes are classified into types: nodes of type x represent those that belong to a particular group, while type y nodes represent those belonging to other groups. For example, type x may represent cat owners and type y non-owners. The selection probability, $\Pi_x(k, t)$ or $\Pi_y(k, t)$, determines which node receives a new edge given its type and current in-degree.

Let $n_x(k, t)$ denote the number of type x nodes with in-degree k at time t , where $k \in \mathbb{Z}$ and $t \in \{0\} \cup \mathbb{Z}^+$, in the DAG constructed as follows:

At each timestep t , one new node, which can be of type x or y , is added to a pre-existing DAG containing type $\{x, y\}$ nodes. The details of how this initial network is constructed are not relevant here and are discussed in Section 3.2.1. A newly added node generates m directed edges, $m \in \{0\} \cup \mathbb{Z}^+$, to a selection of m nodes already in the graph, increasing the in-degree of these selected nodes by one. This process repeats at every timestep, t .

Figure 2 illustrates this in a two-type network with node types a and b . Clearly, any number of types can be reduced to two by labeling one as a focal type as x and all remaining types as y . Throughout this work, examples use this two-type $\{a, b\}$ network: type a nodes play the role of the focal type x , and type b nodes, being non- a , play the role of type y .

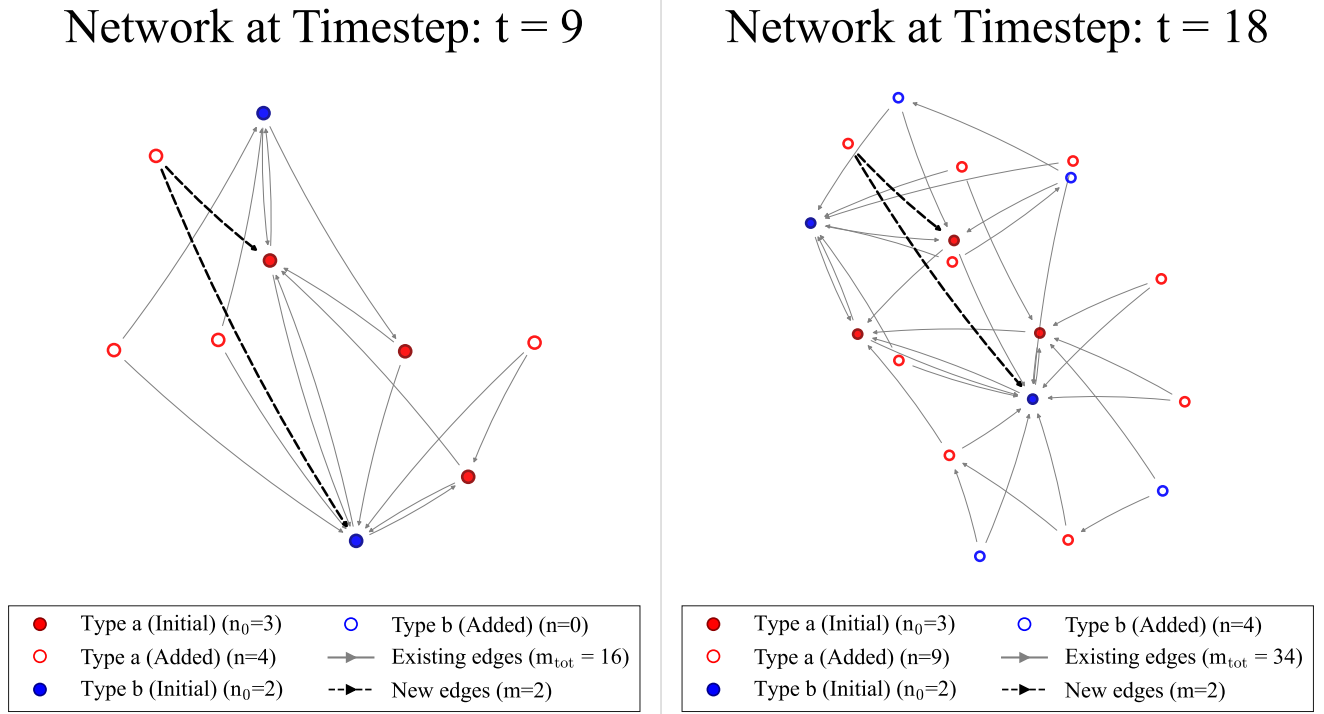


Figure 2: Diagram of the growing directed acyclic graph (DAG) network. Both panels show the same network, which starts as an initial DAG with five nodes (the filled colour nodes). The left panel shows the network at timestep $t = 9$, when four new nodes have been added (the hollow nodes) to the original $n_0 = 5$ nodes. The node added at timestep $t = 9$ has its connections highlighted: the dashed lines indicate that it is in the process of joining the network, and the nodes these dashed arrows point to are those that have been selected probabilistically, with probability $\Pi_a(k, t)$, to receive a connection. The right panel shows the same process at timestep $t = 18$, illustrating how the network has grown further. In graph diagrams, node placement is arbitrary; only the edge set content matters. This is why the positions of the same nodes differ between the two panels.

2.2 Difference Equation

In this model, the expected number of x type nodes with an in-degree k at time $t + 1$ can be expressed as:

$$n_x(k, t + 1) = n_x(k, t) + m\Pi_x(k - 1, t)n_x(k - 1, t) - m\Pi_x(k, t)n_x(k, t) + \delta_{k,0} \quad (3)$$

with the convention that $n_x(k < 0, t) = \Pi_x(k < 0, t) = 0$. This is a difference equation, used as the starting point for deriving the probability that a node of type x has in-degree k at time t [20].

The product term in equation (3),

$$m\Pi_x(k - 1, t)n_x(k - 1, t), \quad (4)$$

is the expected number of nodes that transition from degree $k - 1$ to degree k in a single timestep, t . The quantity $n_x(k - 1, t)$ is the number of these x type nodes with in-degree $k - 1$. The m term is present because the new node adds m edges per timestep, therefore there are m opportunities for

a given type x node to receive a new in-edge. This leaves the $\Pi_x(k-1, t)$ term, which represents the probability that a type x node with degree $k-1$ receives a new in-edge. As an aside, we ignore cases where a newly added node creates multiple in-edges to the same existing node in a single timestep. This is because the terms in equation (3) are approximations and these multiple-edge-to-same-node events constitute an $O(m\Pi)$ correction that becomes negligible in the long-time limit (as $t \rightarrow \infty$) used in this derivation [20]. Now, by the similar reasoning, the term,

$$m\Pi_x(k, t)n_x(k, t), \quad (5)$$

in equation (3) represents the expected number of nodes leaving degree k by gaining an edge and moving to degree $k+1$.

Finally, since we are counting in-degree, the $\delta_{k,0}$ is a Kronecker delta that accounts for the case that, given the newly generated node is type x , it will be an addition to the $n(k=0, t)$ count, as the new node will necessarily have an in-degree of 0.

2.3 Attachment Probability

This derivation extends the Price model [5] by allowing for the attachment probability, $\Pi_{\{x,y\}}(k, t)$, to differ between node types. That is, in general, $\Pi_x(k, t) \neq \Pi_y(k, t)$. We define the node type x probability as,

$$\Pi_x(k, t) = \frac{f_x h_x k + (1 - f_x)(1 - h_x)k + \mu_x}{Z(t)}. \quad (6)$$

Here, f_x denotes the probability that a newly added node is of type x . Since each node is either type x or not, the probability that a node is not of type x is $1 - f_x$.

2.3.1 Homophily and Initial Attractiveness

The parameter h_x denotes a weighting preference that a source node forms a connection with a node of the same type; $h_x \in [0, 1]$. Following Karimi et al., this parameter tunes the degree of homophily in the model [8].

For simplicity, we assign the weighting to a connection between nodes of different types to be $1 - h_x$. This means the homophily parameters, which we can represent by a 2×2 matrix, where the columns i correspond to the type of the newly added node and the rows j to the possible target node types, reduces to,

$$h_{ij} = \begin{bmatrix} h_{xx} & h_{xy} \\ h_{yx} & h_{yy} \end{bmatrix} = \begin{bmatrix} h_x & 1 - h_x \\ 1 - h_x & h_x \end{bmatrix}. \quad (7)$$

This simplification assumes symmetric attachment probabilities for same-type and cross-type pairs. Thus, in equation (6), $(f_x h_x)$ is the expected weighting for same-type nodes of in-degree k to acquire a new edge, while $(1 - f_x)(1 - h_x)$ is the corresponding weighting for the different-type nodes. The consequences of relaxing this assumption are left for future work.

The constant $\mu_x \in \mathbb{R}^+$ is an exogenous parameter controlling the initial attractiveness of nodes of type x , allowing us to study how asymmetric initialisation influences network evolution. For

example, by assigning one node type a substantially larger value of μ , we can investigate the extent of the resulting advantage at a given stage of the network's evolution.

2.3.2 Normalisation Factor

It is useful to recognise that $\Pi(k, t)$ can be rewritten as,

$$\Pi_x(k, t) = \frac{\lambda_x k + \mu_x}{Z(t)}, \quad (8)$$

where we have factorised the k in-degree terms and labeled the resulting factor λ_x ($\lambda_x \in \mathbb{R}^+$). The $Z(t)$ denominator is simply the normalisation factor, ensuring the probabilities of all existing nodes being targeted by the new directed edge sum to 1,

$$\sum_{i \in V_x}^{N_x} \Pi_{x,i} + \sum_{i \in V_y}^{N_y} \Pi_{y,i} = \sum_{i \in V_x}^{N_x} \frac{\lambda_x k + \mu_x}{Z(t)} + \sum_{i \in V_y}^{N_y} \frac{\lambda_y k + \mu_y}{Z(t)} = 1. \quad (9)$$

Using this constraint, we can solve for $Z(t)$.

2.4 Asymptotic Limit

If we conjecture that, as t tends to infinity, both the fraction of nodes of each type and the proportion of in-edges to each type converge to constant values, that is,

$$\lim_{t \rightarrow \infty} E_{\{x,y\}}(t) = \langle g_{\{x,y\}} \rangle t \quad \text{and} \quad \lim_{t \rightarrow \infty} N_{\{x,y\}}(t) = f_{\{x,y\}} t, \quad \langle g_{\{x,y\}} \rangle \in \mathbb{R}^+, \quad (10)$$

we can then rearrange equation (9) for $Z(t)$ in this long-time limit. Here $E_{\{x,y\}}(t)$ denotes the total number of type $\{x, y\}$ in-edges for each respective type, that is $E_{\{x,y\}}(t) \equiv \{E_x(t), E_y(t)\}$. To see this, let us focus on just the N_x contribution to the sum, which we can now rewrite as,

$$\sum_{i \in V_x}^{N_x} \frac{\lambda_x k + \mu_x}{Z(t)} = \frac{\lambda_x \langle g_x \rangle + \mu_x f_x}{Z(t)}. \quad (11)$$

Then, using an identical argument for the y nodes, we can rearrange equation (9) such that,

$$Z(t) = [\lambda_x \langle g_x \rangle + \lambda_y \langle g_y \rangle + \mu_x f_x + \mu_y f_y] t, \quad \text{with rescale } \tilde{Z} \equiv \frac{mt}{Z(t)}. \quad (12)$$

Aside from $\langle g_x \rangle$ and $\langle g_y \rangle$, this is a known analytical expression for $Z(t)$ [20]. The rescaling is purely for algebraic convenience. If we substitute our newly found values into the difference equation (3), we get,

$$\begin{aligned} n_x(k, t+1) - n_x(k, t) = \\ \tilde{Z} t^{-1} [(\lambda_x \langle k \rangle + \mu_x) n_x(k-1, t) - (\lambda_x k + \mu_x) n_x(k, t)] + \delta_{k,0}. \end{aligned} \quad (13)$$

Since we are in the long-time limit, we can rewrite the equation in terms of probabilities by use of,

$$p_x(k, t) = \frac{n_x(k, t)}{N_x(t)}, \quad (14)$$

that is, in this long-time limit, this fraction converges to the probability of the event occurring. This results in,

$$(t+1)p_x(k, t+1) - tp_x(k, t) = \tilde{Z} \left[(\lambda_x(k-1) + \mu_x)p_x(k-1, t) - (\lambda_x(k) + \mu_x)p_x(k, t) \right] + \delta_{k,0} \quad (15)$$

Now, if we further conjecture that each node type has a stable long-time limit probability distribution, i.e.

$$\lim_{t \rightarrow \infty} p_x(k, t) = p_{x,\infty}(k). \quad (16)$$

Then we can rewrite (15) as,

$$\frac{p_{x,\infty}(k)}{p_{x,\infty}(k-1)} = \frac{k + \frac{\mu_x}{\lambda_x} - 1}{k + \frac{\mu_x}{\lambda_x} + \frac{1}{\tilde{Z}\lambda_x}}, \quad (17)$$

where it is useful to motivate, also for algebraic convenience,

$$\alpha_x = \frac{\mu_x}{\lambda_x} \quad \text{and} \quad \gamma_x = 1 + \frac{1}{\tilde{Z}\lambda_x}. \quad (18)$$

In this asymptotic limit, $p_{x,\infty}(k)$ is the probability that a node has an in-degree of k . This is precisely the quantity of interest, which we now isolate using properties of the Gamma function [22].

2.5 Gamma Function Form

If we restrict our domain to $k > 0$, there is a known result that can be used to alter the form of equation (17). It can be written in terms of the ratio of Gamma functions. A gamma function can be defined via the following integral [22]:

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt, \quad z \in \mathbb{C}, \quad \text{where,} \quad \Re(z) > 0. \quad (19)$$

It has the following properties [20],

$$\Gamma(z+1) = z\Gamma(z), \quad \Gamma(1) = 1, \quad \text{defined for all } z \text{ except at } \begin{cases} \Gamma(1-n), & n \in \mathbb{Z}^+, \\ \Gamma(n), & n \in \mathbb{Z}^-. \end{cases} \quad (20)$$

From this definition and its properties, it is possible to rewrite equation (17) as,

$$p_{x,\infty}(k) = \frac{A_x \Gamma(k + \alpha_x)}{\Gamma(k + \alpha_x + \gamma_x)}. \quad (21)$$

Therefore, if we can solve for the constant A_x , we will have our desired result. To do this, we consider equation (15) when $k = 0$. By using our convention that functions with a $((k = 0) - 1)$ term are 0, this results in,

$$p_{\infty,x}(k = 0) \equiv p_{0,x} = \frac{1}{1 + \tilde{Z}\mu_x}. \quad (22)$$

Now, since equation (15) is an iterative equation valid when $k = 0$, we can use it to solve for $p_{x,\infty}(k = 1)$, which is within the domain where equation (21) is a valid. Matching the equation forms and solving for $A_x(k = 1)$, it is found that,

$$A_x(k = 1) = p_{0,x} \frac{\Gamma(\alpha_x + \gamma_x)}{\Gamma(\alpha_x)}, \quad \text{and we conjecture,} \quad A_x = p_{0,x} \frac{\Gamma(\alpha_x + \gamma_x)}{\Gamma(\alpha_x)}, \quad \forall k \in \mathbb{Z}^+. \quad (23)$$

This conjecture is consistent with the known result in equation (21). A full analytical proof is left for future work. However, it can be tested numerically for selected values of k . While the stability of this result requires further analysis over a range of networks, we carried out a brief numerical test. Starting from a network with $n_0 = 5$ initial nodes, we added a further 10,000 nodes. The homophily was set to $h = 0.7$, the probability that a new node is of type a was set to $f_a = 0.6$ and the initial attractivenesses were set to $\mu_a = 1$ and $\mu_b = 2$.

In this setup, the type a and type b node counts deviated from their equation (23) predicted values, $A_x(k)$, by $3.065 \times 10^{-8} \%$ and $-1.065 \times 10^{-8} \%$, respectively, for $k \leq 166$. Within this range, fluctuations around these values were on the order of 10^{-12} . For both types, the computed values $A_a(167)$ and $A_b(167)$, and above, were infinite, likely because the relevant gamma function values could not be evaluated numerically.

The gamma function $\Gamma(z)$ grows rapidly for large arguments, and when z becomes sufficiently large (here $z \approx 167 + \alpha_{x,y} + \gamma_{x,y} \approx 169$), its value exceeds the maximum representable number in the implementation used, causing numerical overflow. Acknowledging the need for future analytical work, we take this numerical analysis as provisional justification for the analytical form of $A_x(k)$, for all $k \in \mathbb{Z}^+$, seen in equation (23).

Now, when representing the A_x found in equation (21) with this equation (23) counterpart, this gives,

$$p_{x,\infty}(k) = p_0 \frac{\Gamma(\alpha_x + \gamma_x)}{\Gamma(\alpha_x)} \frac{\Gamma(k + \alpha_x)}{\Gamma(k + \alpha_x + \gamma_x)}. \quad (24)$$

Whilst this is a perfectly valid final result, we choose to simplify it further. This is done using the Beta function, which can be defined as,

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a + b)}. \quad (25)$$

So, when reintroducing the $k = 0$ case, equation (24) can then be written as,

$$p_{\infty,x}(k) = \begin{cases} p_{0,x}, & k = 0 \\ p_{0,x} \cdot \frac{B(k, \alpha_x + \gamma_x)}{B(k, \alpha_x)}, & k \in \mathbb{Z}^+. \end{cases} \quad (26)$$

This, essentially, concludes the derivation.

2.6 Derivation Closing Remarks

Before moving on to analysing how well the derived equation (26) describes finite-sized realisations of these networks, it is worth highlighting two points. Firstly, given the symmetrical nature of this problem, by simply swapping x for y in equation (26), we ascertain the equivalent equation for the y type nodes. Secondly, the derivation of (26) is not fully analytical.

It is not fully analytical because we lack explicit expressions for $\langle g_x \rangle$ and $\langle g_y \rangle$. These two unknown parameters can however be reduced to one. This is because the total in-degree added at each timestep equals the total out-degree added. This in-out edge conservation necessitates,

$$\langle g_x \rangle + \langle g_y \rangle = m. \quad (27)$$

While deriving analytical expressions for $\langle g_x \rangle$ and $\langle g_y \rangle$ is left for future work, numerically, we observe that the proportion of in-edges per node type do tend to constant values for large t . This can be seen in Figure 3. Moreover, as expected, these constants also sum to m .

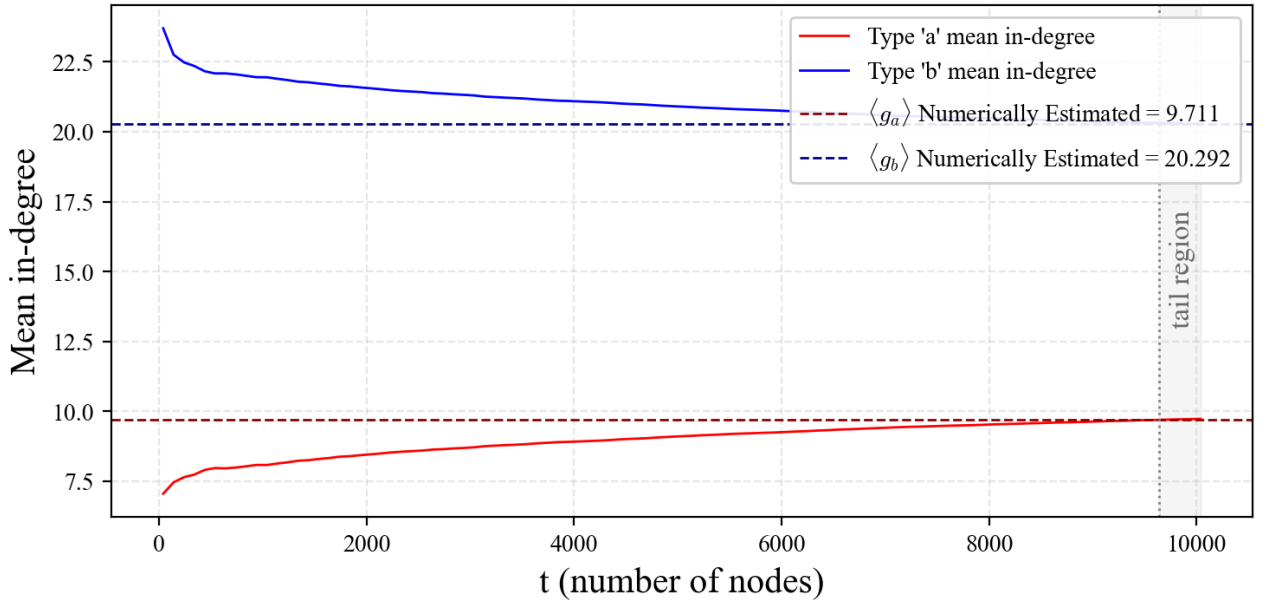


Figure 3: Numerical demonstration of the proportion of in-edges per node type tending to constant values as $t \rightarrow \infty$. This also shows these asymptotes sum to the number of edges added at each timestep, m . The two type $\{a, b\}$ network model has parameters $\{f_a = 0.2, h_a = 0.2, \mu_a = 1, \mu_b = 5, n_0 = 40, N = 10,000, m = 30\}$. In the model, mean values were sampled after every one hundred nodes were added. This value is easily decreased to produce a smaller uncertainty. The final arithmetic mean used for $\langle g_y \rangle$ was then computed as the average of the final five such measurements. The uncertainty between these measurements was on the fractional order 10^{-2} . As seen in Section 6, this network's parameters likely make it poorly described by equation (26); it is included only to illustrate visible convergence to these values. For better-fitting networks (e.g., this one but with $n_0 = 4, m = 3$), the fractional uncertainty decreases to the order of 10^{-5} , and convergence occurs more rapidly. Although $\langle g_x \rangle$ was also calculated numerically and is included here, the value of $\langle g_x \rangle$ used in the model and subsequent statistical analysis was inferred from $\langle g_x \rangle = m - \langle g_y \rangle$.

In practice, we numerically approximate the asymptotic value to which $\langle g_y \rangle$ converges. This is

performed as follows. First, the arithmetic mean in-degree for the type y nodes is measured at regular intervals throughout the simulation as new nodes are added. Then, the arithmetic mean of the measurements taken after at least 95% of the total nodes have been added is computed. The fractional uncertainty is then computed from the standard deviation of these measurements. This mean value is taken to represent the converged behaviour of the data, and hence serves as an estimate of the underlying asymptotic mean, $\langle g_y \rangle$. The corresponding value of $\langle g_x \rangle$ is then obtained from $\langle g_x \rangle = m - \langle g_y \rangle$. The implications of this choice are discussed later in Section 5.5.

This concludes the derivation. The result, equation (26), is a new expression for the probability distribution that a node in a directed acyclic graph with homophily and cumulative advantage (mechanisms known to drive real-world network growth [8]) has an in-degree k at time t .

3 Model Validation

3.1 Approximating Our Distribution as a Power Law

3.1.1 Motivating a Power Law Approximation

One way to assess the accuracy of equation (26) for finite-sized networks is to compare its predictions against data from finite-sized simulations using an appropriate goodness-of-fit test.

Whilst there is modern literature capable of rigorously formulating statistical tests for closely related problems to our own (e.g.[23], [24]), we find that severe difficulties arise when attempting to adapt these approaches to even slightly modified models. In particular, the assumptions required to attain this level of rigour are typically highly technical, tightly constrained, and dependent on asymptotic limit arguments that assume an infinite number of data points. Nevertheless, reviewing this literature helps inform the choices made in Section 5 for the new goodness-of-fit test we ultimately propose.

The majority of this closely related literature is in power law fitting validation. Equation (26), whilst similar, is not a power law. Our function, when $k > 0$, is the ratio of two Beta functions. Equation (26) may be a member of the generalised Waring distribution family however verifying this is left for future work [25]. There is comparatively little statistical validation literature for our distribution contrasted with this substantially richer power law fitting verification literature.

What we now show is, when k is sufficiently large, equation (26) converges to a power law distribution. This approximation turns out to be particularly useful.

3.1.2 Power Law Approximation

As we are considering citations that are positive real numbers, we can make use of Binet's second integral formula, which states, that if the real part of z , $z \in \mathbb{C}$, is positive then,

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + 2 \int_0^\infty \frac{\arctan(t/z)}{e^{2\pi t} - 1} dt. \quad (28)$$

For large enough k and therefore z , the integral term's contribution is negligible, so we drop this term. Now, by exponentiating (28), we have,

$$\Gamma = \sqrt{2\pi} z^{z-1/2} e^{-z}. \quad (29)$$

If we then focus on the numerator and denominator Gamma functions with k arguments in equation (24), and rewrite each of these using equation (29) before recombining the resulting expression, we obtain,

$$p_{x,\infty}(k) \propto \left[\frac{(k + \alpha_x)^{(k+\alpha_x-1/2)}}{(k + \alpha_x + \gamma_x)^{(k+\alpha_x+\gamma_x-1/2)}} \right] e^{\gamma_x}, \quad (30)$$

which can be algebraically manipulated into,

$$p_{x,\infty}(k) \propto \left[\frac{(k + \alpha_x)^{-\gamma_x}}{\left(1 + \frac{\gamma_x}{k + \alpha_x}\right)^{(k + \alpha_x)} \left(1 + \frac{\gamma_x}{k + \alpha_x}\right)^{(\gamma_x - 1/2)}} \right] e^{\gamma_x}. \quad (31)$$

Now, noting,

$$\lim_{k \rightarrow \infty} \left(1 + \frac{\gamma_x}{k + \alpha_x} \right)^{k + \alpha_x} = e^{\gamma_x}, \quad (32)$$

which we can see, for large k , becomes the driving term in the denominator of (31), we therefore have,

$$p_{x,\infty}(k) \propto \frac{(k + \alpha_x)^{-\gamma_x} e^{\gamma_x}}{e^{\gamma_x}} = (k + \alpha_x)^{-\gamma_x}, \quad (33)$$

which is a power law.

3.1.3 Graphical Interpretation of the Approximation

Graphically we can demonstrate this approximation and where exactly it holds well and where it holds less well. Figure 4 demonstrates this for a two-node-type $\{a, b\}$ version of the directed acyclic graph (DAG) model.

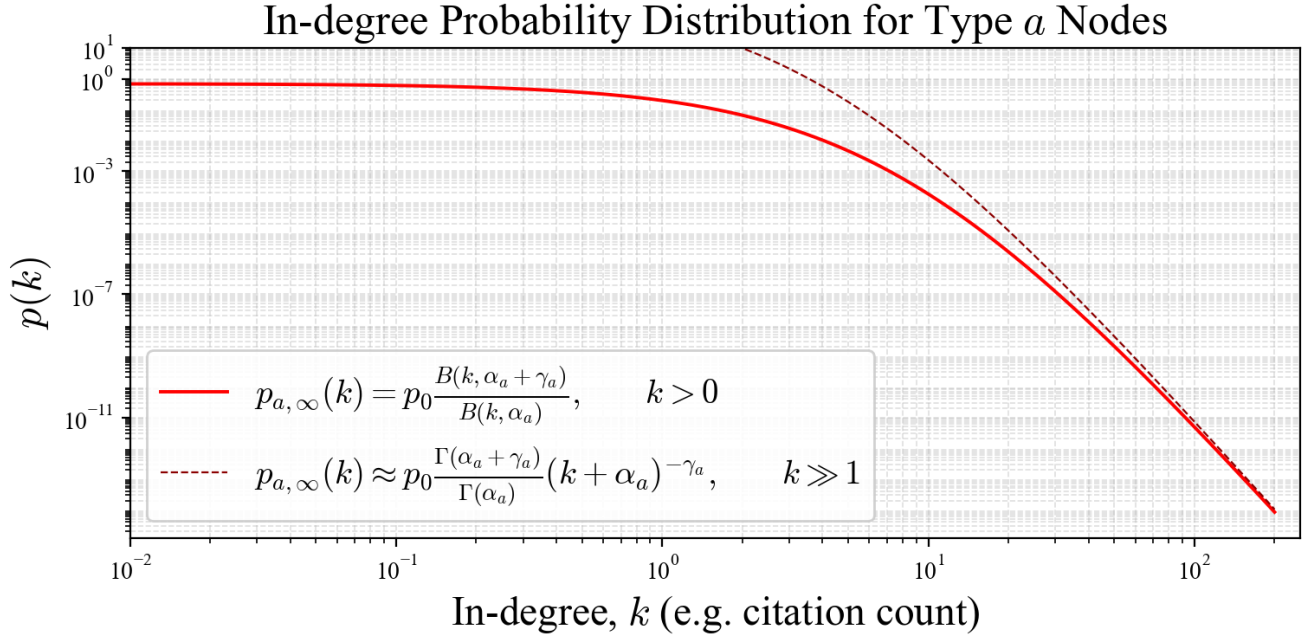
Assessing the accuracy of the approximation graphically is inherently subjective, since deciding when two curves are “approximately the same” is not exactly rigorous. Moreover, the parameter values and network size used in the Figure 4 displayed equation (26) and its large- k approximation, equation (33), are arbitrary.

The network in Figure 4 contains $N_{\text{total}} = 28,004$ nodes. This was chosen only because this is close to the size of the hep-th arXiv dataset, which contains 27,770 papers [13]. The initial network size was set to $n_0 = 4$. This is simply to ensure that it is much smaller than the total network while still permitting $m = 3$. This is because the model requires $m < n_0$, otherwise this would necessitate multiple citations from one paper to another. The remaining parameters were chosen as $h = 0.8$, $f_a = 0.1$, $\mu_a = 1$, and $\mu_b = 5$. This is so that the model exhibits clearly different behaviour for type a and type b nodes, thereby highlighting behaviour captured by equation (26) that is not well described by the conventional literature [5] (corresponding to $h = \{0, 0.5, 1\}$, $f_a = \{0, 0.5, 1\}$ and $\mu_a = \mu_b$ in the language of our new derivation).

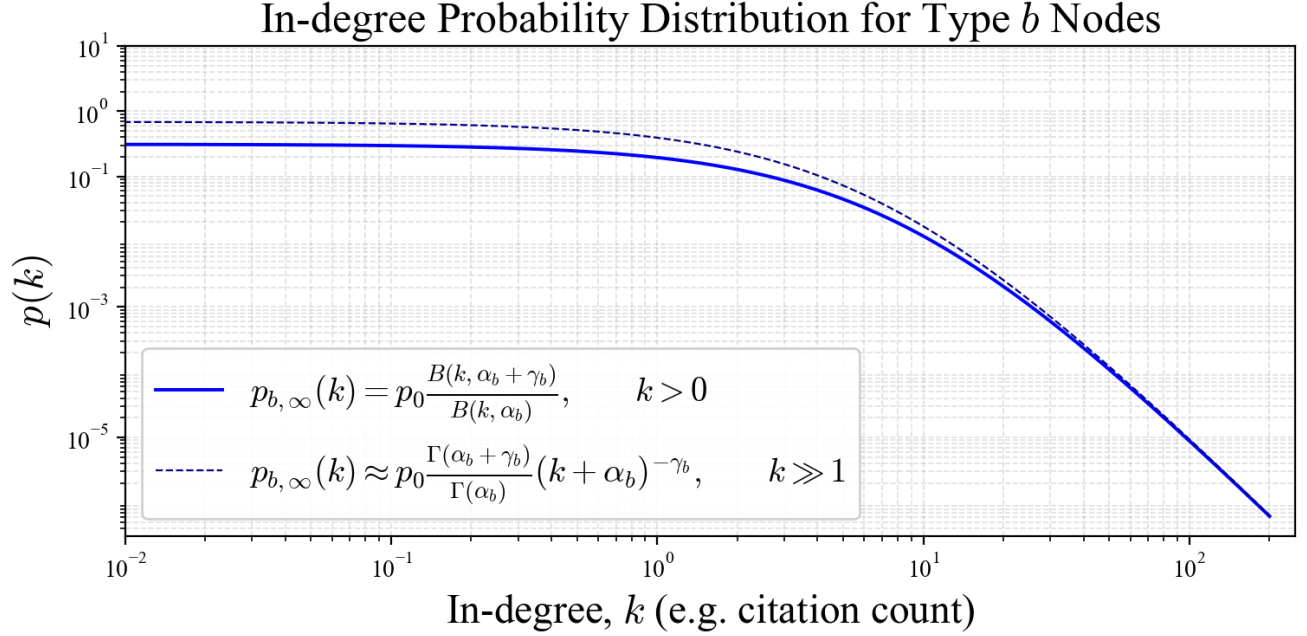
Changing the network parameters changes the model behaviour. Since the purpose of the theory is to model real networks, where such quantities are fixed by the real data and may vary widely between cases, graphical comparison is useful only for illustrating specific examples such as Figure 4, and can be limited as a more general measure of the model’s goodness-of-fit.

A further limitation arises when adding simulated or empirical degree data to plots like Figure 4. In network science, such data are commonly displayed using logarithmic histogram bins, where each bin is larger than the previous one by a fixed factor, typically around 1.1 [20]. The representative horizontal coordinate of each logarithmic bin is often taken to be its geometric mean [20], for example,

$$k_{\text{bin}} = \sqrt{k_{\text{min}} k_{\text{max}}}. \quad (34)$$



(a) Sub-figure plotting equation 26 (solid curve) and equation 33 (dashed curve) for node type a .



(b) Sub-figure plotting equation 26 (solid curve) and equation 33 (dashed curve) for node type b .

Figure 4: Theoretical in-degree distribution for a two type $\{a, b\}$ version of the directed acyclic graph (DAG) network outlined in Section 2. The network contains $N_{\text{total}} = 28,004$ nodes ($n_0 = 4$, $N = 28,000$), $m = 3$ out-edges per node, homophily parameter $h = 0.8$, type a fraction $f_a = 0.1$, and type-specific attachment rates $\mu_a = 1$, $\mu_b = 5$. The solid curves show the exact asymptotic distribution (equation 26), while the dashed curves show the power law approximation (equation 33) derived from Binet's second integral formula when $k \gg 1$. Both distributions are plotted on log-log axes to reveal over what orders of magnitude the approximation is accurate.

This introduces additional uncertainty through the arbitrary choices involved in a given binning scheme, making it unclear whether apparent deviations arise from the model itself or from these choices made in the plotting procedure.

For these reasons, this work uses graphs only to illustrate selected examples, while the main analysis is left to the now further motivated original goodness-of-fit method developed in Section 5 that is less restricted by these drawbacks. This is particularly important because, at large k , genuine deviations from both equations (33) and (26) occur because of finite-size effects, which we now discuss.

3.2 Initial Conditions

3.2.1 Finite-size Effects and Initial Conditions

As discussed, no numerical model can realise an infinite number of nodes, and consequently any convergence arguments taken in the limit as t tends to infinity must be interpreted as approximations. Deviations between finite simulated networks and their infinite-size theoretical counterparts are called finite-size effects [26]. Ultimately, our work is motivated by the desire to more accurately describe finite-sized, real-world citation networks, therefore, careful treatment of these finite-size effects is required. We study these finite-size effects empirically by analysing how the initial conditions of our simulation influence the resulting finite networks.

Finite-size effects in network models are significantly influenced by the choice of the initial conditions we set up our simulation with [27]. For some initialisations, deviations from a given model persist even in the infinite-size limit [27]. Berset and Medo quantify this effect for a similar network model to ours, the Barabási–Albert preferential attachment model [4, 27]. This model can be thought of as describing the exact same directed acyclic graph (DAG) as the one we study, however, for a single type network with an initial attractiveness equal to zero (i.e. $f_x = \{0, 1\}$ and $\mu = 0$). In the special case where each new node introduces a single edge (i.e. $m = 1$), the probability that a randomly selected node has degree k in the infinite-size limit is given by:

$$p(k) = \frac{4}{k(k+1)(k+2)}, \quad (35)$$

[28]. Berset and Medo show that this result can accurately describe a network only when the nodes in the initial DAG to which Barabási–Albert nodes are added have an average in-degree below 3. This result persists even in an infinitely sized network. Above an average initial network in-degree of 3, this small advantage allows for the attraction of so many in-edges that these initial nodes never converge to the stationary degree distribution of later added nodes. In other words, the highest in-degree nodes in the model deviate from the predicted in-degree distribution.

We conjecture that a similar conclusion holds for our network: as the size and edge density (the fraction of possible edges that are present [29]) of the initial network increases, equation (26) provides a poorer description of the resulting finite-sized network that grows from it. This poor fit is driven by the behaviour of the high in-degree nodes. We further expect this to hold for all values of m , not just $m = 1$. In Section 5, we provide a numerical goodness-of-fit procedure allowing us to test this, while a theoretical proof is left for future work. In Section 6 we perform this test.

3.2.2 Initialisation Choices

Since we want to investigate real-world networks, we want an initial network that balances mitigating these finite-size effects with producing structures that resemble real-world patterns. We now investigate this balance. We term the initial conditions we consider in this study,

3.2.2.1. Minimally intrusive initialisation.

3.2.2.2. Variable edge density initialisation.

Initially in this study we considered initialisation (3.2.2.1). Here, the idea being that a potentially suitable initialisation is one with a topology resembling that produced by the network growth process we wish to model.

To do this, we initialise the model with a fraction f_x of x type nodes. For a probabilistically constructed model, the initial network contains n_0 nodes with an expected,

$$\mathbb{E}[n_{0,x}] = f_x n_0 \quad \text{and} \quad \mathbb{E}[n_{0,y}] = f_y n_0. \quad (36)$$

The next step is to randomly select a source node, which then connects to a target node chosen according to a specified probability distribution. For example, if we let the x type nodes have a probability of forming a connection with a node another node in the initial system,

$$\Pi_x(k, t) = \frac{\lambda_x k + \mu_x}{Y(t)}, \quad t \in t_{\text{initial}}, \quad (37)$$

then to find $Y(t)$ we follow a derivation similar to that of $Z(t)$ (12) found earlier. There are however two key differences. Firstly, since the source node is already a member of the network, we need to explicitly specify that the node cannot form a connection with itself. Secondly, we cannot use the asymptotic convergence trick used in equation (10) to find a linear constant. This is because the in-degree per node-type is non-constant in $t \in t_{\text{initial}}$. To account for this first discrepancy, we introduce an additional step in the algorithm that checks the type and in-degree of the randomly selected node and removes the contribution it would otherwise make to $Y(t)$. For example, if the randomly selected node is of type x and has total in-degree k_i (where i labels the node under consideration) at time $t \in t_{\text{initial}}$, then $Y(t)$ needs to be rewritten as,

$$Y(t) = \lambda_x (E_x - k_i) + (n_{0,x} - 1)\mu_x + n_{0,y}\mu_y, \quad (38)$$

where a reasonable choice for the total edge counts is,

$$E_{\text{total}}^{(\text{in})} = \mathbb{E}[E_x(t)] + \mathbb{E}[E_y(t)] = \lambda_x m t + \lambda_y m t, \quad t \in t_{\text{initial}}. \quad (39)$$

However, this (3.2.2.1) initialisation has a drawback: since $t_{\text{initial}} < \infty$, the realised values will not necessarily match the expected ones. Alternatively, using a deterministic initialisation that matches the expected values exactly would restrict us to parameters that generate integer expectations. This constraint would severely restrict the model's application potential. We therefore conclude that an initialisation resembling that of our network either fails to align perfectly with the model,

thereby allowing the initial nodes to likely retain an unfair advantage that is non-convergent with our model, or is limited in its range of applicability.

This brings us to the network initialisation choice (3.2.2.2). Extrapolating from the Berset-Medo result, we have conjectured that the initial network’s size and edge density shapes finite-size effects in our model. How large are these effects, and at which initial sizes and edge densities do they matter? The answer tells us when our theory is informative for real networks and when finite-size effects obscure it. Because an analytical solution is difficult, we seek a statistic to compare across the number of edges, m , per initial node, and the number of initial nodes, n_0 . With a well-chosen statistic, we can empirically identify initial edge densities that balance mitigating finite-size effects with producing a model suitable for analysing real-world networks. To proceed, we need an initial condition in which the number of edges, or out-degree, each node makes (m) and the initial number of nodes (n_0) can vary. This what initialisation choice (3.2.2.2) gives us.

The procedure to set up (3.2.2.2) is identical to (3.2.2.1), except that the constraints enforcing the homophilic weighting mirroring are removed. Once the correct fraction is set and the rule preventing self-connections is enforced, all we do is allow ourselves the ability to assign different values to m and n_0 . With these conditions specified, we now discuss our choice of statistic used for comparison.

4 Fitting Our Model

4.1 Goodness-of-Fit Difficulties for Networks

To understand for which initial conditions equation (26) provides a reasonable description of our artificial citation network, we make use of a goodness-of-fit analysis. As a reminder, goodness-of-fit measures how well a statistical model describes a set of observations. A natural starting point is the familiar goodness-of-fit measure: the χ^2 statistic [30, 20],

$$\chi^2(\theta_j; y_i) = \sum_{i=1}^N \left[\frac{y_i - f(x_i; \{\theta_j\})}{\sigma_i} \right]^2, \quad (40)$$

where y_i is the observed values at point i , $f(x_i; \{\theta_j\})$ is the theoretical prediction evaluated at x_i given the set of parameter values $\{\theta_j\}$, and σ_i^2 is the variance associated with data point i . In many physics applications, χ^2 offers a compact and intuitive summary of how well a theoretical distribution matches observed data.

However, the citation networks we consider are heavy-tailed. A heavy-tailed distribution is one where there is a larger probability of getting very large values compared to exponential decay [20]. For heavy-tailed power law distributions, several of the usual assumptions underlying χ^2 -based inference can fail [24]. The heavy-tailed power law distribution,

$$f(x) \propto \frac{1}{x^\gamma}, \quad x \in \mathbb{R}^+, \quad (41)$$

where normalisation implies $\gamma > 1$, is the exact form of equation (33). Since we observe that equation (26) converges to (33), we can heuristically conclude that equation (26) is, at least in the domain where the two equations are similar, heavy-tailed.

At this point, one may ask: *'If both equations (26) and (33) concentrate a high proportion of data at the beginning of the distribution, away from the high in-degree tail, is it not overly pedantic to abandon χ^2 and other standard tests merely to account for an essentially negligible set of outliers?'* To answer this question, we recall from section 3.2.1 that we conjecture these outliers represent the region where the model is most likely to deviate. Consequently, a goodness-of-fit analysis for equation (26) that can capture tail behaviour is crucial. Moreover, this is precisely the region where our power law approximation holds, which explains its relevance to this analysis.

For $f(x)$ (41), one can show that if $\gamma \leq 2$, no finite moments (no mean, no variance, etc) exist [31], while for $2 < \gamma \leq 3$ the variance diverges and the central limit theorem, in its classic formulation, does not apply, instead the mean is normally distributed with an infinite standard deviation [24]. It is important to caveat that these results apply to power law distributions on an infinite domain. However, although our finite network simulations necessarily have bounded degrees and therefore finite empirical moments, the asymptotic regime from which both (26) and (33) are derived assumes an infinite domain; one in which the second moment may diverge. Consequently, asymptotic approximations that rely on finite second moments and independent sampling, such as the classical χ^2 distribution, would not be theoretically justified.

This difficulty is further compounded by the dependence between the nodes in our model induced by the attachment probability given by equation (6). An analysis of the extent to which the nodes in our model are dependent, that is, whether they may be regarded as weakly dependent, thereby

opening avenues for alternative statistical test, is left for future work. Moreover, the standard SciPy χ^2 implementation assumes Poisson distribution-approximated bin counts, an assumption that does not hold for our network.

4.2 Fitting Power Laws Debate

Clearly, for our citation network, we require a goodness-of-fit test that is not impaired by the problematic features outlined in Section 4.1. Although the need for such a test is widely acknowledged, there remains active debate over whether an appropriate method currently exists in the literature [19, 32, 23, 18]. Our review of this literature indicates the need for a new goodness-of-fit procedure to adequately assess how well equation (26) describes our network. Before introducing the test we propose, it is useful to outline the statistical concepts on which it relies by discussing a closely related procedure: the influential test of Clauset et al., *Power law distributions in empirical data* [18].

The procedure described in Clauset et al. [18] provides a method for generating a p -value when comparing empirical data with a fitted power law model. In its given form, this test is also inappropriate for evaluating the goodness-of-fit of equation (26) on our network. To both make this limitation clear and to introduce most of the statistical machinery used in our Section 5.6 method, we now review the Clauset et al. procedure.

4.3 Clauset et al. Method

The Clauset et al. method begins by considering a set of real empirical data and a power law $f(x)$ (equation (41)). The aim is to assess the goodness-of-fit of $f(x)$ in describing the data. The subscript n is a variable that indicates the size, n of the sample. Let the domain of $f(x)$ (the values for which $f(x)$ is defined) be $a \leq x_i \leq b$ where $a \in \mathbb{R}^+$ and $b \rightarrow \infty$. The a parameter is then varied within its domain and, for each variation a Kolmogorov-Smirnov (KS) [33] statistic is computed.

4.3.1 KS Test

We can think of a KS test as a statistical test that provides a quantitative indication of how likely it is that two continuous, one-dimensional probability distributions come from a given reference probability distribution. The resulting test is a computation of the following statistic,

$$D_n = \sup_x |F_n(x) - F(x)|, \quad (42)$$

where $\sup_x$ is the supremum of the set of distances. The supremum of the set of distances is defined as the largest value attained when taking $|F_n(x) - F(x)|, \forall x$, or if no largest value is attained, the smallest number that is still greater than or equal to all of them [34]. Intuitively, all this means is the statistic takes the largest absolute difference between the two distributions across all x values.

The distribution $F_n(x)$ is the empirical cumulative distribution function (CDF),

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n 1_{(-\infty, x]}(X_i), \quad \text{where} \quad 1_{(-\infty, x]}(X_i) = \begin{cases} 1 & \text{if } X_i \leq x \\ 0 & \text{otherwise.} \end{cases} \quad (43)$$

X_i denotes the i -th observed data value in a sample of size n . $F(x)$ is the CDF for the theoretical distribution supposing it is continuous (which of course in our case it isn't. This discrepancy is discussed further in Section 5.2),

$$F(x) = \int_{-\infty}^x f(t) dt \quad (44)$$

where $f(x)$ is the continuous analytical probability distribution. Clearly, before the KS statistic can be computed, one must first obtain $F_n(x)$ and $F(x)$. While $F_n(x)$ can be computed directly from the data, the evaluation of $F(x)$ requires parameter estimation. To this end, as we now discuss, Clauset et al. employ maximum likelihood estimation.

4.3.2 Maximum Likelihood Estimation

Maximum likelihood estimation (MLE) treats observed independent and identically distributed (i.i.d.) random variables as drawn from the same distribution with given parameters $\{\theta_j\}$ [35, 20]. The plausibility of a given parameter choice is quantified by a likelihood function. Under the i.i.d. assumption, this likelihood can be written as a product of the individual probabilities of each observation, $P_\theta(x_i)$:

$$L(\theta) = \prod_{i=1}^n P_\theta(x_i). \quad (45)$$

The larger this value, the larger the likelihood our data can be considered to have been drawn from this function. Given that the natural logarithm is monotonic, maximizing $L(\theta)$ is equivalent to maximizing the log-likelihood. So given,

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log P_\theta(x_i), \quad \text{we search for } \{\theta_j\} \text{ such that, } \nabla_\theta \ell(\theta) = 0, \quad (46)$$

where the solutions of the gradient equation are the MLE derived parameters.

At this point, the relevance of MLE to our analysis may be questioned by asking: *'If the in-degree of a given node, predicted by $p_{x,\infty}(k)$ (26), is not independent across nodes, how can MLE be applied to fit the parameters of $F(x)$, when, in this case, the probability density is $p_{x,\infty}(k)$?'*

Fully addressing the tension between MLE's i.i.d. assumption and our network's inherent dependencies would require a rigorous analytical treatment of weak dependence conditions. However, we can proceed heuristically on the following grounds. The data of interest are those that deviate. We conjecture these are the high-degree nodes, deviating as a result of the initial conditions. A sufficiently large dataset will place the in-degree of these nodes in the regime where equation (26) converges to a power law, (33). This matches the form of $f(x)$ (41). Clauset et al. demonstrate empirically that MLE fitting of $f(x)$ produces reliable parameter estimates [18]. This justifies, for sufficiently large datasets, the future use of MLE for our analysis.

So, for a given value of a , Clauset et al fit a power law model, $f(x)$ (41), to the truncated empirical data with $x \geq a$ using MLE. The resulting MLE-fitted power law defines a new theoretical CDF,

$F_{[a,\infty)}(x)$ normalised over this truncated $[a, \infty)$ domain. The KS statistic is then computed between this MLE-fitted CDF, $F_{[a,\infty)}(x)$, and the corresponding empirical CDF, $F_{n \in [a,\infty)}(x)$, which is also normalised over the truncated domain.

The next step is to then choose the a value from the set of a values computed that generates the smallest KS statistic for this MLE fitted empirical data. We refer to this particular a as a^* and this corresponding smallest KS statistic in the set of a values as $D_{n,\text{emp}}(a^*)$. The subscript emp is shorthand for empirical to indicate the resulting KS statistic comes from the MLE-fitted power law, which is based on empirical data.

4.3.3 Bootstrapped p-value

With a value for $D_{n,\text{emp}}(a^*)$ in hand, Clauset et al. [18] repeat the same fitting and testing procedure on many synthetic data sets generated via a semi-parametric bootstrapping algorithm. For observations below $x_{\min}$, the smallest observed data value greater than or equal to a , they resample data from the empirical distribution; this is referred to as non-parametric bootstrapping. We will not use non-parametric bootstrapping in our method, hence, the details of this procedure are omitted in this paper. However, for observations above $x_{\min}$, multiple sets of synthetic data are generated from the fitted power law model via a Monte-Carlo process; this is referred to as parametric bootstrapping. This procedure allows one to produce the set,

$$D_{n,\text{syn}} = \{D_{n,\text{syn},1}(a^*) \dots D_{n,\text{syn},N}(a^*)\}, \quad i = \{1, \dots N\}, \quad (47)$$

from which they compute an empirical p -value,

$$p = \frac{\#\{D_{n,\text{syn},i}(a^*) > D_{n,\text{emp}}(a^*)\}}{|\{D_{n,\text{syn},i}(a^*)\}|}, \quad (48)$$

where $\#\{D_{n,\text{syn},i}(a^*) > D_{n,\text{emp}}(a^*)\}$ is the number of synthetic samples for which this inequality holds. The subscript syn is shorthand for synthetic to indicate the corresponding resulting KS statistics come from the synthetically generated data. This concludes their method.

4.3.4 Closing Remarks on Clauset et al.

Within Corral and Deluca's *Fitting and goodness-of-fit test of non-truncated and truncated power law distributions*, they note that Clauset et al. provide no explanation of why their method should work [19]. Corral and Deluca, however, do make conjectures that begin to answer this question.

Their argument stems from the fact that, in principle, the distribution of the KS statistic is known, at least asymptotically, independently of the underlying form of the distribution, $F(x)$ [19],

$$p = 2 \sum_{j=1}^{\infty} (-1)^{j-1} \exp \left[-2j^2 (D_{n,\text{emp}} \sqrt{N} + 0.12 D_{n,\text{emp}} + 0.11 D_{n,\text{emp}} / \sqrt{N})^2 \right]. \quad (49)$$

Since $F(x)$ is absent in equation (49), we refer to this equation (49) as distribution-free. This result however relies on convergence of the empirical process to a mathematical object called a Brownian bridge [23], which requires both i.i.d. observations and a continuous, fully specified, distribution [23]. As discussed in Section 4.1, our citation network violates these conditions, so equation (49) is

not formally valid. Furthermore, because the MLE step re-estimates both γ_{emp} and γ_{syn} based on the data observed, the KS statistic is biased. Nonetheless, Corral and Deluca show that equation (49), when computed via the ‘probks’ routine (an algorithm developed by Press [36]) behaves as an upper bound on Clauset et al.’s equation (48) while retaining a reasonably similar functional form [19]. From this, they argue, if the data follows a power law, increasing N (by lowering a) should reduce the KS statistic, since the distribution free equation (49) implies $d_e \sim N^{-1/2}$. If, however, lowering a introduces a departure from power law behaviour in the data, the resulting model misclassification may increase d_e . This argument however relies on the misclassification bias outweighing the variance reduction from the larger sample. Corral and Deluca suggest there is no general reason to think this.

Furthermore, it is not explicitly stated in Clauset et al. why (48) constitutes a p-value. Therefore, we put forward the following heuristic argument for why (48) can be considered as one. A p -value can be defined as,

$$p = \int_{\text{rejection region}} \rho(x) dx, \quad (50)$$

where $\rho(x)$ is a probability density function, defined such that the infinitesimal probability of finding the outcome in the range x to $x + dx$ is $\rho(x) dx$ [30]. In our setting, if this integral is approximated by a Monte Carlo sum over synthetic data sets, we can then estimate the probability mass in the rejection region as the fraction of samples where $D_{n,\text{syn},i}(a^*) > D_{n,\text{emp}}(a^*)$. This is exactly Clauset et al.’s equation (48), hence why Clauset et al. refer to it as an empirical p-value.

Finally, within, *A practical recipe to fit discrete power law distributions* [37], Corral demonstrates that if one uses a fully parametric procedure, data truly generated from a power law distribution is rejected less frequently in comparison with Clauset et al.’s semi-parametric bootstrapping. After fitting the model via MLE over the selected data range, synthetic datasets are generated entirely from the fitted distribution, with appropriate normalisation over the truncated domain to ensure that the probability mass sums to one. The KS statistic is then recomputed for each synthetic dataset to obtain a more accurate result from their empirical p-value (48).

5 Our New Method

5.1 Motivation For New Goodness-of-Fit Method

Whilst the method of Clauset et al. is widely used in power law modelling literature [18, 19, 37], we argue, that it cannot be applied to our setting without modification. Nor can we apply the goodness-of-fit method of Corral and Deluca [19] to our particular problem. We now explain why these methods cannot be applied directly and introduce a new procedure to address this limitation. Our method allows us to compute a heuristic goodness-of-fit metric for how well a heavy-tailed distribution matches a dataset with non-independent data points. With this method we can assess the goodness-of-fit of equation (26) for finite-sized simulations.

5.2 Discrete Data Correction

Our data is discrete; citation counts take integer values. This means the classical continuous KS distribution, equation (42), does not apply. As shown rigorously by Dimitrova et al. [32], when the theoretical function being investigated is discrete, the KS statistic is no longer distribution-free: its distribution depends explicitly on the underlying CDF, $F(x)$. They find, when using values derived from this continuous KS test, that the p-values produced are too large. We concede that whether this leads to inflation in the p-values produced by our method remains an open question.

What we can do however, for damage limitation, is ensure that our theoretical function is discrete. This is done in Corral et al. [37]. Here, they use a discrete power law modelled using a Hurwitz zeta function expressed numerically via a Euler-Maclaurin expansion up a chosen cutoff as opposed to the continuous power law equation (41) [38]. Naturally, they find, using a discrete power law to model discrete phenomena produces more accurate results from their goodness-of-fit test compared to using the continuous power law, $f(x)$ (41). Thankfully, equation (26) is already defined on the same set as our citation data, the positive integers, so, no correction to it is required.

5.3 Truncating From Above

What we are interested in is how well the asymptotic limit derived equation (26) describes our finite-sized simulated data. To assess this, we employ the upper truncation procedure described by Corral and Deluca [19]. Specifically, for a fixed lower bound a , we progressively decrease the upper truncation threshold b_i , restricting the data to the interval $[a, b_i]$. For each such interval, the model parameters are re-estimated via MLE, and a goodness-of-fit test is performed using a fully parametric bootstrap.

The aim is to determine the largest interval $[a, b^*]$ on which the model cannot be rejected at a prescribed threshold p-value equal to some chosen p_c . Formally,

$$b^* = \max(b_i) \quad \text{s.t.} \quad p([a, b^*]) \geq p_c \quad (51)$$

where $p([a, b_i])$ denotes the bootstrap p -value computed under the model truncated to $[a, b_i]$.

The choice of p_c therefore acts as a model-selection threshold. It governs the maximal domain of non-rejection. Corral suggests setting a safe p_c value of $p_c = 0.2$ [19]. However, as we can see

from the arbitrary height below which we define the rejection, this choice, as with all significance values, is relatively arbitrary. For this reason, in Section 6, we produce our analysis for a variety of p_c values and examine the sensitivity of our results to this choice.

Furthermore, it is important to state explicitly that as b_i decreases, both the empirical and bootstrap KS statistics are computed over a reduced domain. In turn, the procedure should be interpreted as identifying the largest domain on which the model remains statistically consistent with the data, rather than as establishing a global goodness-of-fit.

Based on our analysis in Section 3.2.1, we conjecture that the size of the domain for which $p([a, b^*]) \geq p_c$ depends on the extent to which the initial conditions and finite-size effects distort the highest in-degree nodes. In particular, if deviations from equation (26) are concentrated in the extreme tail, we expect upper truncation to reveal a breakdown of the model in this region.

5.4 KS Reweighting

Beset and Medo demonstrate that, when a data sample is large enough, the initial network is dense and the initial network exceeds a certain small size, the reweighted KS statistic is more suitable for statistical analysis of a truncated power law distribution [27]. So, as opposed using equation (42), we use the reweighted,

$$D_n = \frac{\sup_x |F_n(x) - F(x)|}{\sqrt{F(x)(1 - F(x))}} \quad (52)$$

Clauset et al. note that the KS statistic is known to be relatively insensitive to differences between distributions at the extreme limits of the range of x . This is because in these limits the CDFs, $F_n(x)$ and $F(x)$, necessarily tend to zero and one.

Clearly, in studying the impact of finite-size effects on the high-degree nodes, it is crucial our analysis is sensitive in the CDF regions where they approach one. In using the reweighted KS test we avoid this problem as the reweighted KS test is uniformly sensitive test across all $[a, b^*]$ ranges. Thankfully, Clauset et al. note, for their method, the reweighted KS performs very similarly to the standard KS statistic [18].

5.5 Inference Problem Reframing

Another way in which the aim of our analysis differs from [19, 37, 18] is that we are not asking whether empirical data are well described by a given theoretical distribution (power laws in their case, equation (26) in ours). Rather, we ask whether a distribution derived strictly in an asymptotic limit provides an adequate description of a subset of data generated by a finite-sized simulation.

This requires a subtle re-framing of the inferential problem. For this re-framing it is important to revisit the caveat in our section 2 derivation that we have not derived a second analytic relation alongside equation (27) that would allow us to solve explicitly for $\langle g_x \rangle$ and $\langle g_y \rangle$, nor do we estimate these quantities via MLE. Instead, we take the values of $\langle g_x \rangle$ and $\langle g_y \rangle$ obtained from the first network instantiation and treat them as if they were the ones derived from analytic theory.

Conditional on the assumption that these initial values may be treated analogously to MLE-derived parameters, we then simulate Monte Carlo realisations and perform MLE fitting for p_0 , γ_x , and α_x (and the corresponding y parameters when reversing subject type). Under this interpretation,

the structure of the goodness-of-fit procedure remains formally similar to that of the Clauset et al. method [18], even though the underlying inference objective differs.

5.6 Recipe For New Goodness-of-Fit Method

We now provide our recipe for the computation of a heuristic goodness-of-fit metric for how well a heavy-tailed distribution matches a dataset with non-independent data points. With this method we can assess the goodness-of-fit of equation (26) for finite-sized simulations. This will conclude our goodness-of-fit derivation.

Choose an arbitrary cutoff of p-value, p_c . For example, $p_c = 0.2$.

Let the domain of the data over which we assess the goodness-of-fit of equation (26) be $a \leq x_i \leq b$, where $a, b \in \{0\} \cup \mathbb{R}^+$. Throughout our analysis we fix $a = 0$, although, as shown in [19], it may in principle also be varied. Select a value b_i such that $\{b_i\} \subseteq [a, x_{max}]$, and let the largest element of this set be $b_{max} = x_{max}$, the largest simulated data point. This choice ensures that the first test is performed on the full dataset before any upper truncation is imposed.

In principle, for each b_i , the reweighted KS statistic (52) is computed. However, as we will see, it is often unnecessary to perform this computation for every interval $[a, b_i]$.

Replace $F(x)$ (44) with the appropriate corresponding expression. In our case, the integral form of equation (44) associated with equation (26) is not analytically solvable. We therefore compute it numerically. After normalising over the truncation range $[a, b_i]$ containing k values ranging from k_{min} to k_{max} , equation (44) becomes,

$$F_{x,\infty}(k) = \frac{\sum_{i=k_{min}}^{\min(k, k_{max})} p_{x,\infty}(k_i)}{\sum_{i=k_{min}}^{k_{max}} p_{x,\infty}(k_i)} \quad (53)$$

For $F_n(x)$, following [37], we use the empirical survival function as opposed to empirical CDF,

$$F_n(x) = \frac{N_x}{N_{\{a,b\}}}, \quad (54)$$

where N_x denotes the number of observations in the truncated sample satisfying $x_i \geq x$, and $N_{\{a,b\}}$ is the total number of data points in the range $a \leq x_i \leq b$. Now, using equation (52), this results in the a set of KS statistics for each b_i , $D_{n,theory}([a, b_i])$.

Using the same data as before, we repeat the procedure to generate $F_n(x)$, but with one key change: instead of using the theoretical distribution from equation (26), we now use the same distribution but this time MLE fitted to our actual data. Specifically, we estimate the distribution parameters using MLE applied to the data in $[a, b_i]$. This gives us a new set of KS statistics for each interval, which we denote $D_{n,MLE}([a, b_i])$.

For each data instantiation, we collect both KS statistics into a set:

$$D_j = \{D_{n,theory,j}([a, b_i]), D_{n,MLE,j}([a, b_i])\} \quad (55)$$

where $j = 1, \dots, N$ indexes the data instantiation. We repeat this procedure for N instantiations of the data.

We then compute the empirical p -value using equation (48), starting with the largest value of b_i (x_{max}) and working down in order of size.

Once we find a value b^* :

$$b^* = \max(b_i) \quad \text{such that} \quad p([a, b^*]) \geq p_c \quad \text{and} \quad b^* \in \{b_i\} \subseteq [a, x_{max}], \quad (56)$$

the p -value computed for the data in $[a, b^*]$ provides sufficient evidence that the data are consistent with the theoretical distribution. Since we search from largest to smallest domain, we stop at this first such domain. If no satisfactory domain is found across all b_i , then we conclude the data cannot be reasonably described by the distribution. This concludes the algorithm.

We now discuss the results of applying this algorithm to our heavy-tailed, asymptotic-limit derived probability distribution (26) to quantify how well it describes subsets of data generated by finite-sized simulations. In other words, we now possess a tool for determining when and where equation (26) can reliably describe finite-sized networks.

6 Results

6.1 Motivating Initial Network Effect Contour Plots

Real-world networks vary widely in size and topology (e.g. [4, 9, 16]), and the models used to represent them therefore span a wide range of parameter values. Our goodness-of-fit test can be used to identify which parameters significantly affect how accurately equation (26) describes a finite-sized network’s observed in-degree distribution.

As a reminder, the directed acyclic network model derived in Section 2 contains parameters $n_0, m, N, f_x, h, \mu_x, \mu_y$. It begins with an initial network of n_0 nodes, each forming m directed edges to other initial nodes, and then grows by sequentially adding N further nodes according to equation 6, each of which also forms m directed edges to existing nodes. Each new node is of type x with probability f_x or not-of-type x with probability $1 - f_x$. The parameter h controls homophily, while μ_x and μ_y are the type-specific initial-attractiveness parameters.

As a first application of the goodness-of-fit framework, we examine how the validity of equation (26) varies with the initial network. We therefore vary m and n_0 , which control the number of edges per node and the size of the initial seed network, while fixing f_x, h, μ_x, μ_y and N . This choice is guided by the discussion in Section 3.2.1, where initial network size and edge density were identified as a plausible source of model deviation from the simulated networks.

For each fixed (m, n_0) network, the goodness-of-fit procedure in Section 5.6 produces a maximal accepted cut-off in degree, b^* , for a chosen threshold p -value, p_c . Rather than reporting b^* directly, we express the result as the fraction of the network retained below this cut-off. This gives a more comparable summary across networks: one network may be well described up to $b^* = 20$ and another up to $b^* = 40$, but if their in-degree ranges differ substantially these raw cut-offs are difficult to compare directly. We therefore compute, for each (m, n_0) network, both the fraction of nodes retained and the fraction of in-edges retained including and below b^* . Figures 5 and 6 show these quantities as contour plots over the (m, n_0) plane, with colour indicating the retained fraction.

Each plotted (m, n_0) network uses $m \in \{1, \dots, 34\}$, sampled at every other value, and $n_0 \in \{10, \dots, 150\}$, sampled every ten values. We fix N at two values, $N = 10,000$ (left-hand columns) and $N = 1,000$ (right-hand columns), to compare a larger and a smaller network. The larger and more computationally expensive value is of the same order as the High Energy Physics Theory (hep-th) arXiv dataset (containing 27,770 papers [13]). Other parameters were fixed at $h = 0.2$, $f_a = 0.2$, $\mu_a = 1$, and $\mu_b = 5$ to induce distinct type a and type b behaviour, highlighting effects captured by equation (26) but not necessarily by the conventional literature [5].

For the goodness-of-fit test, the per-network data were split into 50 equally spaced citation ranges, and truncation was applied at each. Although logarithmic spacing could in principle better reflect the uneven in-degree distribution predicted by equation (26), we chose uniform spacing as to avoid overly broad windows in the data sparse high-degree tail. For each truncation, 50 simulated networks were generated as synthetic data following Section 5.6, and compared against one original simulated network treated as the network under analysis. Future work should assess the sensitivity of the results depending on the choice of truncation scheme.

6.1.1 Edge Fraction Retained

Figure 5 shows that the retained edge-fraction is broadly stable across the three values of p_c , suggesting that the results are not highly sensitive to the cutoff choice. For both $N = 1,000$ and $N = 10,000$, the highest retained fractions occur in the bottom-left of the (m, n_0) plane, indicating that equation (26) performs best for small and relatively sparse initial networks.

For $N = 1,000$, variation across the majority of the admissible region is modest: beyond the well-performing bottom-left corner, much of the grid shows similar behaviour, with a noticeable deterioration only in the highest-density region just above $m = n_0$ for $m \gtrsim 10$. For $N = 10,000$, the contrast is much sharper: the poor-fit bottom-right region spreads outward from the highest-density corner, reducing the retained edge-fraction for previously reasonably-well-fit networks, whereas networks outside this region now satisfy the fit criterion over a larger edge-fraction. Increasing N therefore does not simply improve or worsen the fit uniformly, but makes the distinction between favourable and unfavourable regions of parameter space more pronounced. Given the complex dependence of network behaviour on model parameters, this comparison should be regarded as preliminary. A systematic analysis of the role of N is clearly important future work.

6.1.2 Node Fraction Retained

Figure 6 shows that the retained node-fraction is also broadly stable across the three values of p_c , again suggesting limited sensitivity to the cutoff choice.

For $N = 1,000$, the retained node-fraction is close to one across almost the entire admissible grid, with noticeable deterioration only in the same region (as with the edge-fraction for $N = 1,000$) just above $m = n_0$ for $m \gtrsim 10$. In other words, even where the retained edge-fraction falls, equation (26) still describes the majority of the nodes in the majority of these networks well. This indicates that, for networks of this size, when the edge-level fit is poor, this is driven mainly by a relatively small set of highly connected nodes.

For $N = 10,000$, the picture changes substantially. The node-fraction plot becomes much more similar to its edge-fraction counterpart: that is, the lower-right of the admissible region becomes a broad poor-fit zone, while the left-hand side of the grid remains comparatively well described. In this regime, the breakdown of equation (26) is therefore no longer confined to a small number of influential nodes, but extends across a far larger fraction of the network.

6.1.3 Figures Taken Together

Taken together, Figures 5 and 6 provide a clearer picture of model validity than either does alone. The edge-fraction plots show where a substantial proportion of in-edges are poorly described by equation (26), while the node-fraction plots show whether this failure is concentrated among a small number of influential nodes or spread across a larger fraction of each network.

Moreover, across Figures 5 and 6, two competing effects can be seen at work. Larger networks should better approximate the asymptotic relations, (10), underlying the theory, which should improve agreement with equation (26). At the same time, larger networks allow the unexpectedly high-degree nodes more opportunity to accumulate in-edges, potentially worsening the fit. Again, determining how these effects balance requires analysis considering far more values of N .

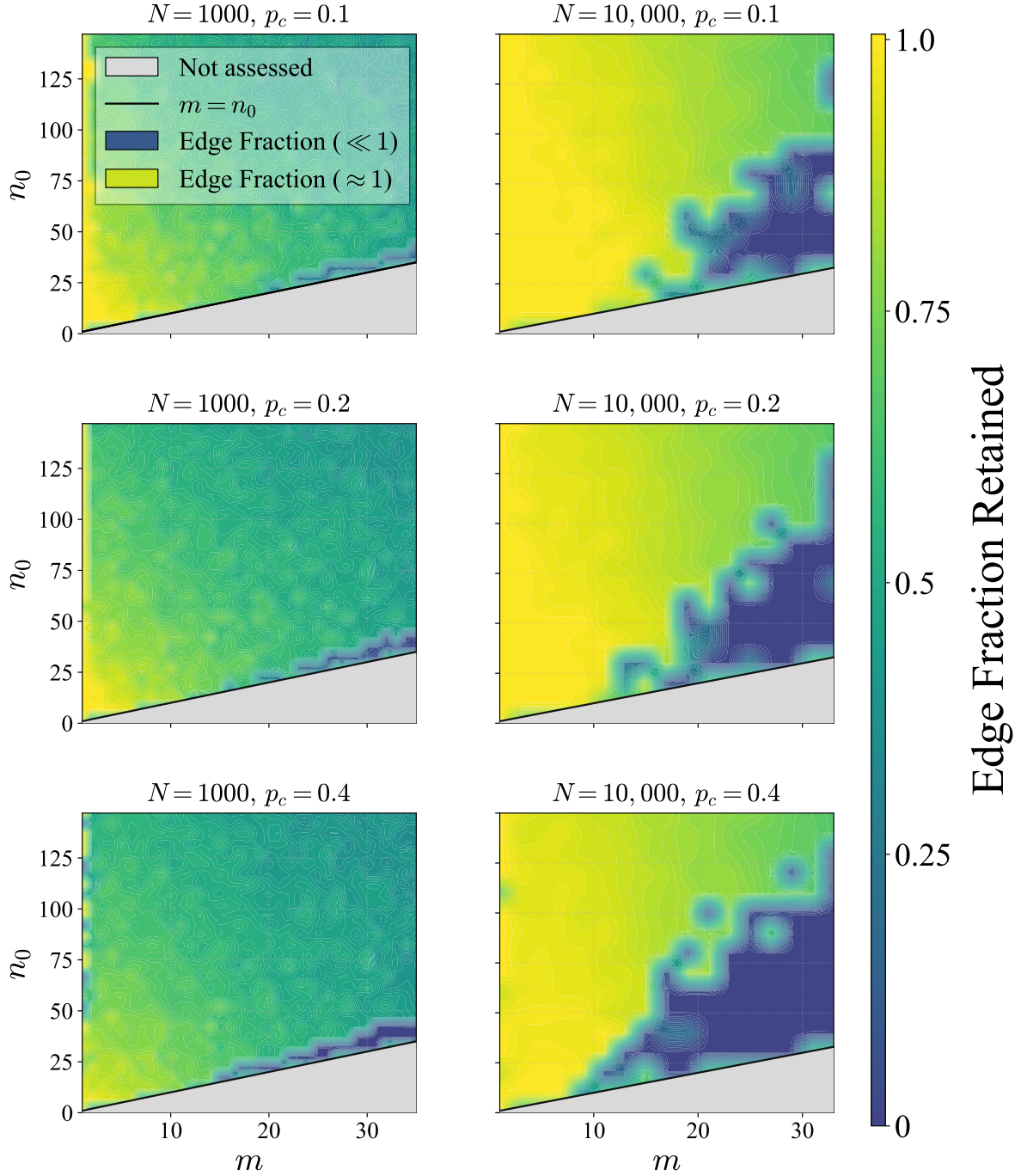


Figure 5: Contour plots displaying the fraction of edges for which our in-degree probability distribution (26) produces p -values exceeding threshold p_c in simulated finite-sized networks, assessed via our goodness-of-fit test (Section 5.6). Six subplots are organized in a 3×2 grid varying: (i) $p_c \in \{0.1, 0.2, 0.4\}$ (rows) and (ii) total nodes added $N \in \{1,000, 10,000\}$ (columns). Networks were generated by sequentially adding N nodes with attachment probability $\Pi_x(k, t)$ (equation 6) to initial graphs constructed via variable-edge-density initialisation (Section 3.2.2.2). Each grid point is parameterised by m (out-degree per node, horizontal axis) and n_0 (initial graph node count, vertical axis). The colour scale represents the fraction of edges satisfying the p -value threshold (dark blue = 0; bright yellow = 1). Grey regions indicate the clearly network-growth-algorithm breaking parameter combinations where $m > n_0$. Results shown for type b nodes in $\{a, b\}$ type networks. The network and goodness-of-fit parameters are given in Section 6.1.

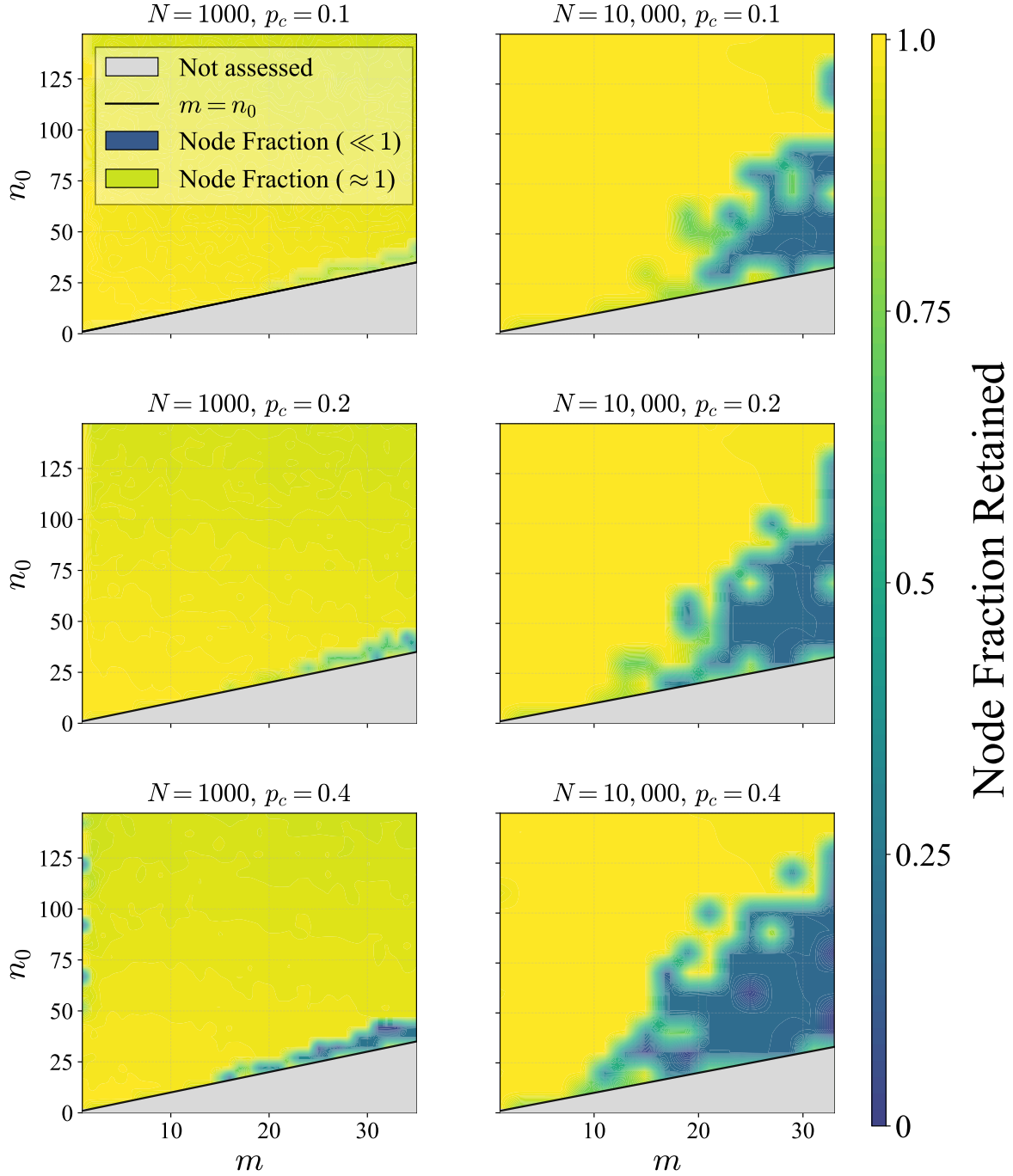


Figure 6: Contour plots displaying the fraction of nodes for which our in-degree probability distribution (26) produces p -values exceeding threshold p_c in simulated finite-sized networks, assessed via our goodness-of-fit test (Section 5.6). Six subplots are organized in a 3×2 grid varying: (i) $p_c \in \{0.1, 0.2, 0.4\}$ (rows) and (ii) total nodes added $N \in \{1,000, 10,000\}$ (columns). Networks were generated by sequentially adding N nodes with attachment probability $\Pi_x(k, t)$ (equation 6) to initial graphs constructed via variable-edge-density initialisation (Section 3.2.2.2). Each grid point is parameterised by m (out-degree per node, horizontal axis) and n_0 (initial graph node count, vertical axis). The colour scale represents the fraction of nodes satisfying the p -value threshold (dark blue = 0; bright yellow = 1). Grey regions indicate the clearly network-growth-algorithm breaking parameter combinations where $m > n_0$. Results shown for type b nodes in $\{a, b\}$ type networks. The network and goodness-of-fit parameters are given in Section 6.1.

7 Concluding Remarks

This work develops both a new in-degree prediction and corresponding goodness-of-fit procedure for networks in which cumulative advantage [5] and homophily [8, 7] are the key mechanisms dictating network growth. In particular, the proposed goodness-of-fit test (see Section 5.6) makes it possible to identify which regions of a finite network are reliably described by the asymptotic-limit-derived equation (26).

Although developed for the present model, the same procedure could in principle be extended to related network models for in-degree probability distributions, including those of Price [5], Barabási–Albert [4], and Karimi et al. [8]. This raises the possibility of using our derived goodness-of-fit framework, valid in principle across these distributions, to assess which model is most appropriate for in-degree prediction in different real-world networks [4, 9, 8]. Conversely, one could hold equation (26) fixed and compare our new goodness-of-fit procedure with existing alternatives, such as the widely used but often contested test of Clauset et al. [18, 19].

In the present analysis, attention has been restricted to varying m , the number of out-edges per node, and n_0 , the size of the initial network. A natural next step would be to vary the number of nodes added, N . This would help disentangle two competing effects: increasing N should improve agreement with the asymptotic theory, conversely however, larger networks allow unexpectedly high-degree nodes more opportunity to accumulate in-edges, therefore potentially worsening the fit. An (N, m) contour plot of the retained node and edge-fractions could provide significant insight into the balance between these effects.

A further extension would be to vary the lower cutoff a , rather than fixing it at zero. This would test whether the largest retained fraction always lies on the low-degree side, and would clarify whether high-degree nodes generally drive the misfit across networks of varying topologies.

Finally, the derivation presented here is not yet fully analytical, since at present either $\langle g_x \rangle$ or $\langle g_y \rangle$ must still be determined numerically (see Section 2.6). Obtaining a fully analytical treatment therefore remains an important direction for future work.

Taken together, these results provide both a new theoretical description of in-degree in networks whose growth is governed by cumulative advantage [5] and homophily [8, 7], and a practical framework for determining where and when that description can be used reliably in finite-sized networks.

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